

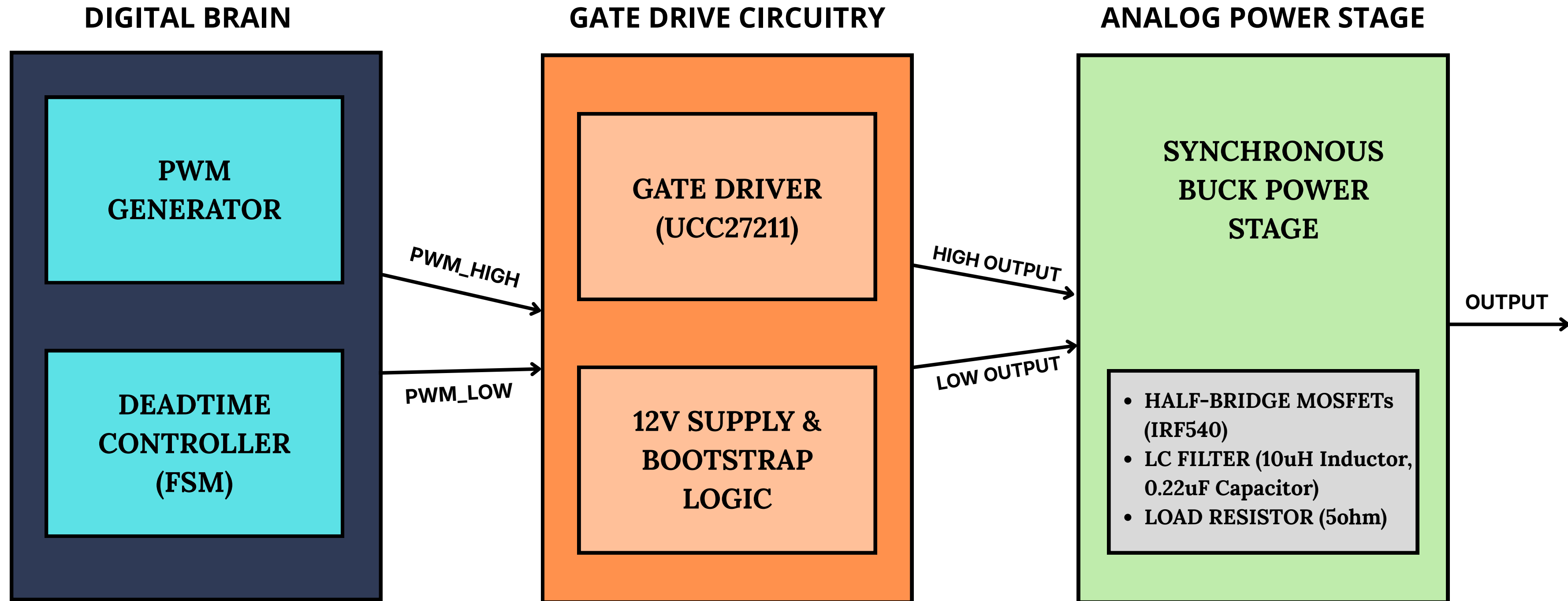
Design & Co-Simulation of a Mixed-Signal Synchronous Buck Converter

“Integration of Digital Verilog Control with an Analog Power Stage”

TOOLS:

- **Analog Co-Simulation: Cadence PSpice**
- **Digital Logic Design: Verilog HDL**
- **RTL Verification: Icarus Verilog & Gtkwave**
- **Code Editor: Visual Studio Code**

BLOCK DIAGRAM



PWM GENERATION

+

DEADTIME CONTROLLER

→

MAIN CONTROLLER

pwm_gen.v

```
1 module pwm_gen(pwm_out , clk , rst , duty);
2     parameter period = 100 ;
3
4     input clk , rst ;
5     input [7:0] duty ;
6     output reg pwm_out ;
7
8     reg [7:0]count ;
9     reg [7:0]duty_reg ;
10
11     always @(posedge clk , negedge rst) begin
12
13     if(!rst) begin
14         count <= 0 ;
15         pwm_out <= 0 ;
16         duty_reg <=0 ;
17
18     end else begin
19         if(count<period)
20             count <= count+1 ;
21         else begin
22             count <=0 ;
23             duty_reg <= duty ;
24
25         end
26
27         pwm_out <= (count<duty_reg)?1:0 ;
28
29     end
30
31 end
32
33 endmodule
```

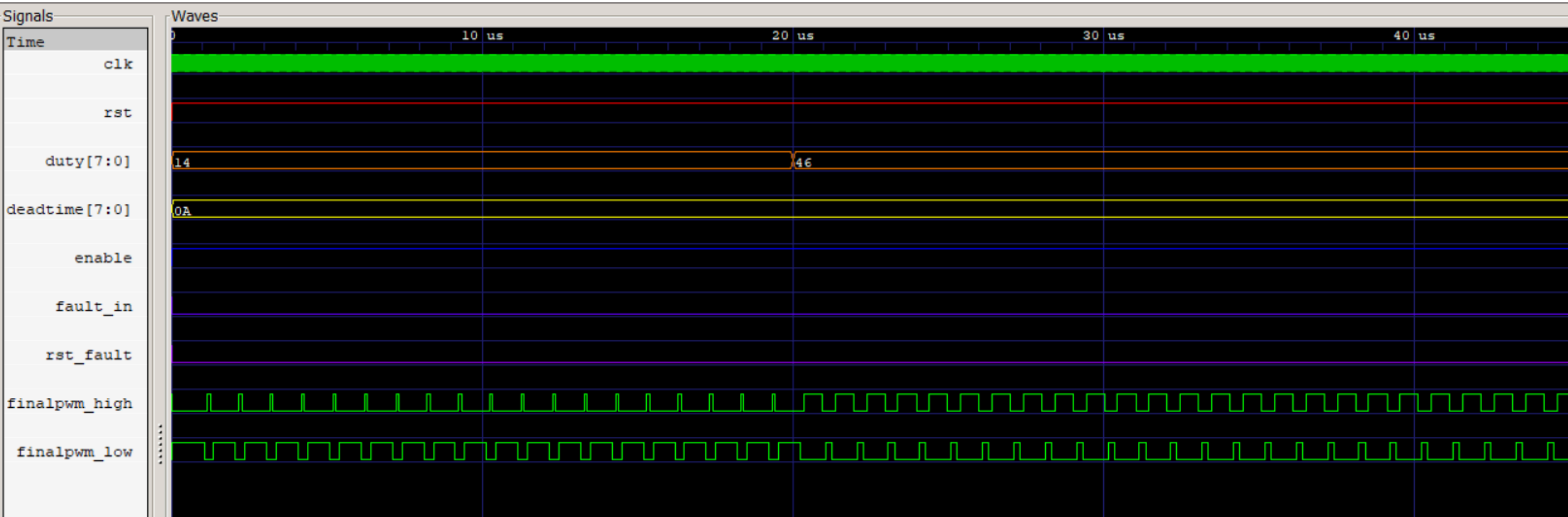
deadtime.v

```
1 module deadtime(pwm_high , pwm_low , clk , rst , master_pwm
2     input clk , rst , master_pwm ;
3     input [7:0]deadtime ;
4     output reg pwm_high , pwm_low ;
5
6     localparam state_low_on = 2'b00 ;
7     localparam state_deadtime_rise = 2'b01 ;
8     localparam state_high_on = 2'b10 ;
9     localparam state_deadtime_fall = 2'b11 ;
10
11     reg [1:0]current_state ;
12     reg [7:0]count ;
13
14     always @(posedge clk , negedge rst) begin
15         if (!rst) begin
16             pwm_low <=0 ;
17             pwm_high <=0;
18             count <=0 ;
19             current_state <= state_low_on ;
20         end
21         else begin
22             if(current_state==state_low_on) begin
23                 pwm_low <= 1;
24                 pwm_high <=0;
25                 count <=0;
26                 if (master_pwm) begin
27                     current_state <= state_deadtime_rise ;
28                 end
29             end
30
31             if(current_state==state_deadtime_rise) begin
32                 pwm_low <=0;
33                 pwm_high <=0;
34                 if(count<deadtime) begin
35                     count <= count+1 ;
36                 end
```

top_gatedriver.v

```
1 module top_gatedriver(finalpwm_high , finalpwm_low , cl
2     input clk , rst ;
3     input [7:0]deadtime_value ;
4     input [7:0]duty_cycle ;
5
6     input enable , fault_in , rst_fault ;
7
8     output finalpwm_low ;
9     output finalpwm_high ;
10
11     wire finalmaster_pwm ;
12     wire wirepwm_high ;
13     wire wirepwm_low ;
14
15     reg fault_state ;
16
17     always @(posedge clk , negedge rst) begin
18         if(!rst) begin
19             fault_state <= 1'b0 ;
20         end
21         else if (rst_fault==1'b1) begin
22             fault_state <= 1'b0 ;
23         end
24         else if (fault_in==1'b1) begin
25             fault_state <= 1'b1 ;
26         end
27     end
28
29
30     pwm_gen master_pwm_gen (
31         .clk(clk),
32         .rst(rst),
33         .duty(duty_cycle),
34         .pwm_out(finalmaster_pwm)
35     );
36
37     deadtime deadtime_gen (
```

Verilog Simulation Waveform



```
enable=0; fault_in=0; rst_fault=0; duty=0; deadtime=0;
rst = 0; #20;
rst = 1;

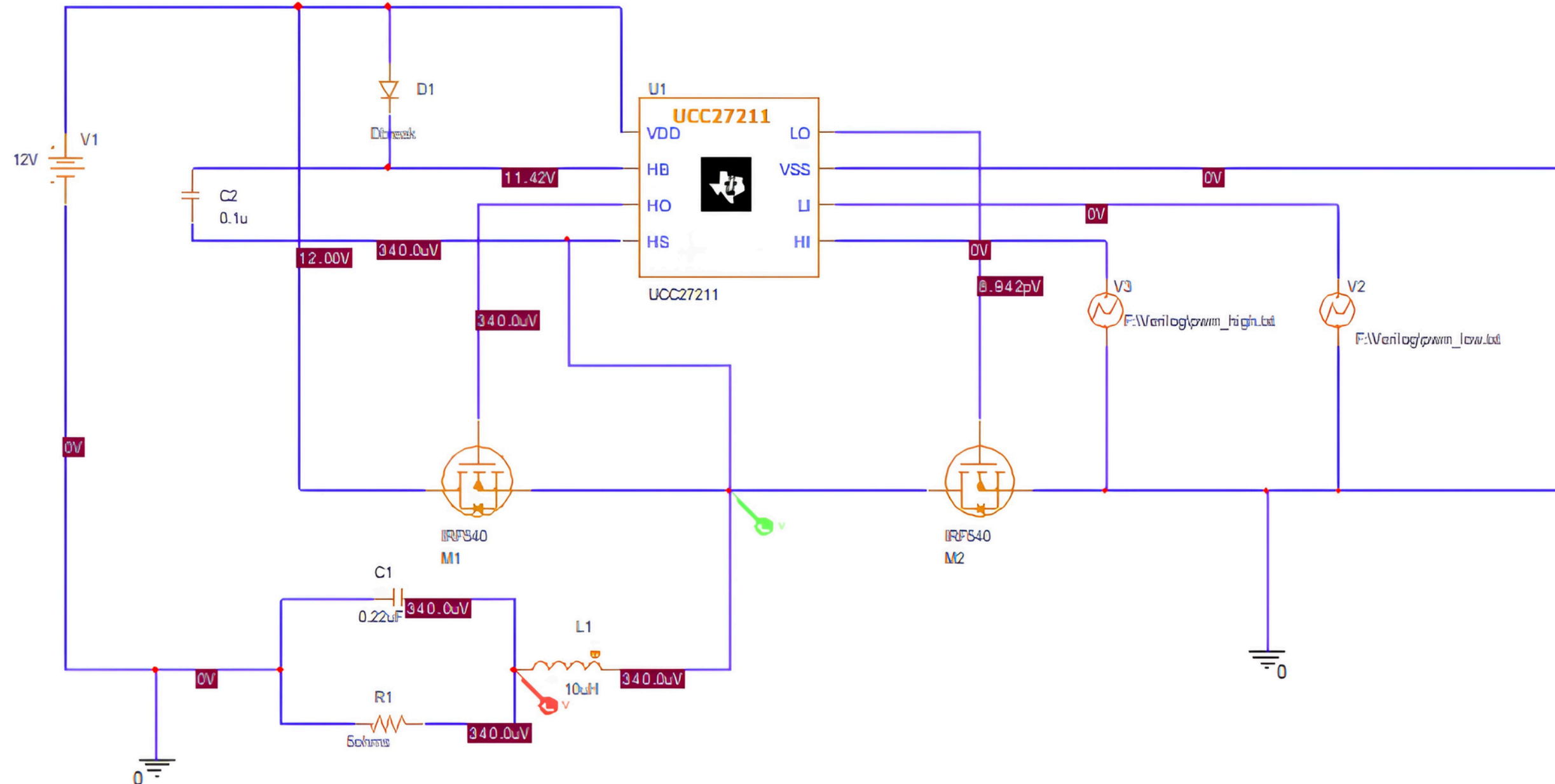
// Low Voltage
duty = 20;
deadtime = 10;
enable = 1;
#20000;

// High Voltage
duty = 70;
#25000;

$finish;
```

TESTBENCH SNIPPETS

- Duty cycle instantly and flawlessly shifts from 20% to 70% at exactly 20μs.
- pwm_high & pwm_low signals did not overlap due to deadtime logic
- Zero Shoot-Through which confirms absolute protection for the analog hardware.

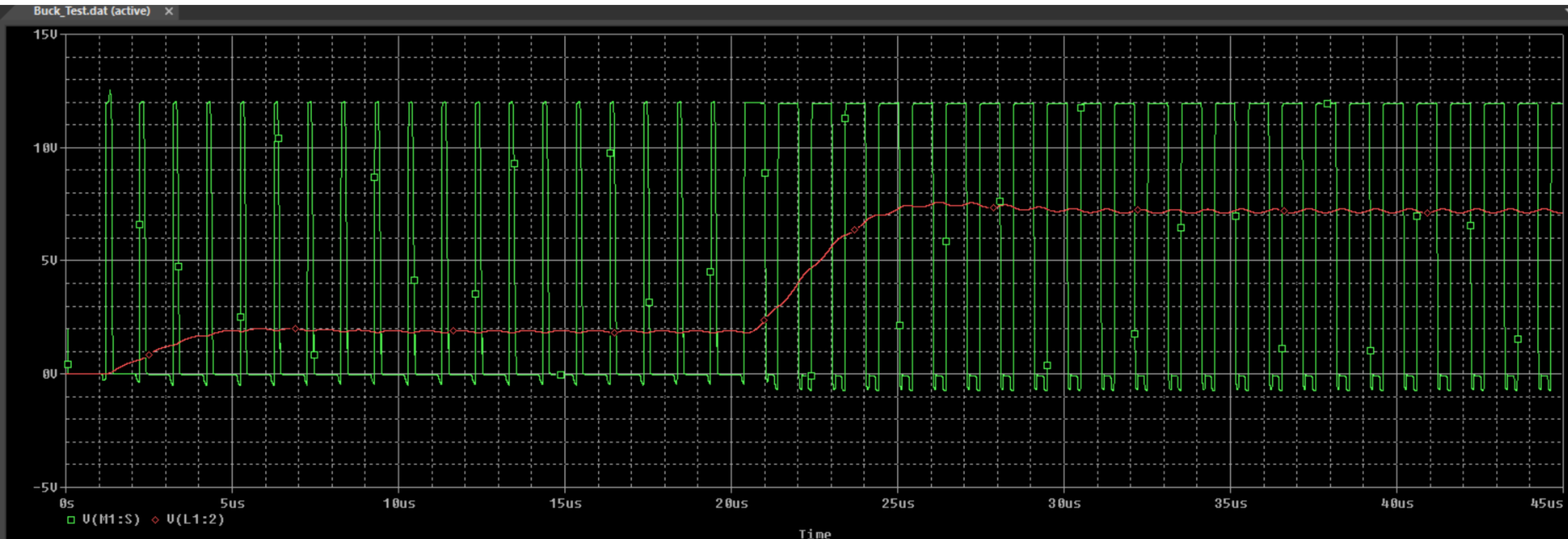


BUCK CONVERTER CIRCUIT DIAGRAM

COMPONENTS

- Gate Driver IC: Texas Instruments UCC27211
- Power Switches: 2 × IRF540 (N-Channel MOSFETs)
- Bootstrap Circuit: 0.1μF Capacitor (C2) & Power Diode (D1)
- Underdamped LC Filter: 10μH Inductor (L1) & 0.22μF Capacitor (C1)
- Output Load: 5Ω Resistor (R1)
- Digital Stimulus (VPWL_FILE) -V2 & V3

Transient Analysis & Step-Response Verification



- **Green Trace (Switching Node):** Verilog FSM flawlessly shifts the duty cycle from 20% to 70% at exactly 20 μs .
- **Red Trace (Output Voltage):** Shows a smooth step-response, transitioning the DC output from 2.4V to a stable 8.4V